

V4.2 Toy Model 3G

Quantum Dynamics Close-Out and Guardrail Consolidation

Rev A - Consolidated Close-Out Paper for Toy Models 3A-3F

Date: 2026.05.25 | Continuation after Toy Model 3F

Status: This close-out consolidates the quantum-dynamics track. It records structural compatibility results only. It does not claim a derivation of the Standard Model, quantum field theory, particle masses, charges, spin values, or physical 3T existence.

1. Abstract

Toy Model 3 explored whether the quantum-dynamics machinery used in standard quantum mechanics can be represented as the projected structure of hidden transverse temporal phase geometry. The track began with complex amplitudes as real rotations in a hidden phase plane, then extended through Schrodinger-type time evolution, spatial momentum, kinetic energy, scalar potentials, electromagnetic gauge coupling, and the relativistic Klein-Gordon / Dirac bridge.

The consolidated result is that the 3T → 1T reciprocal projection architecture remains mathematically compatible with the central formal structures of quantum dynamics. The work is not a proof that the proposed 6D architecture is physically real. It is a controlled sequence of structural compatibility tests.

2. Executive conclusion

Toy Model 3 shows that complex amplitudes, unitary time evolution, free-particle wave dynamics, scalar potentials, gauge connections, and relativistic spinor structure can all be represented as successive constraints on hidden transverse phase geometry after a 3T → 1T projection.

The most defensible statement is:

The 6D reciprocal projection framework provides a coherent geometric representation of standard quantum-dynamics structures, but it does not yet derive the constants, source equations, particle spectrum, or experimental predictions required for a physical theory.

3. Toy Model 3 progression

Model	Target	Main result	Guardrail
3A	Complex amplitudes	A real transverse phase plane with $J^2 = -I$ is algebraically equivalent to complex phase.	Does not prove complex numbers are unavoidable or derive the Born rule.
3B	Time dynamics	Uniform hidden phase rotation gives Schrodinger-type energy-eigenstate evolution.	Does not derive the physical Hamiltonian H.
3C	Spatial waves / kinetic term	Spatial phase twisting gives momentum and kinetic-operator structure.	Uses standard dispersion; does not derive mass or $\hbar$.
3D	Scalar potentials	$V(x)$ acts as a position-dependent hidden phase-rotation-rate shift.	Does not derive real force fields.
3E	Gauge connection / EM coupling	Local phase freedom requires connection fields A and Φ , matching minimal coupling.	Does not derive Maxwell equations, charge, or QED.
3F	Relativistic bridge	Relativistic first-order evolution forces matrix factorization and spinor structure.	Does not derive particle masses, spin values, QFT, or the Standard Model.

4. Toy Model 3A - Complex amplitudes from hidden phase geometry

Toy Model 3A started from real geometry only. A hidden 3T space was written as:

$T_{\text{hidden}} = \text{span}(\tau_0, \tau_1, \tau_2)$

The observable 1T projection selects τ_0 . The two remaining hidden temporal directions form a transverse phase plane:

$$P_{\text{transverse}} = \text{span}(\tau_1, \tau_2)$$

On that plane, the real 90-degree rotation operator is:

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$J^2 = -I$$

This is the structural bridge to the imaginary unit i . Multiplication by i also means a 90-degree rotation and also satisfies $i^2 = -1$. Therefore a complex amplitude $z = u + i v$ can be represented as a real vector (u, v) in the hidden transverse phase plane.

$$\begin{aligned} R(\phi) &= \cos(\phi) I + \sin(\phi) J \\ R(\phi)(u, v) &\leftrightarrow \exp(i \phi) z \\ |z|^2 &= u^2 + v^2 \end{aligned}$$

3A gives a geometric representation candidate for complex amplitudes. It does not derive the Born rule and does not prove physical 3T existence.

5. Toy Model 3B - Dynamics from phase rotation

Toy Model 3B made phase dynamic. Uniform hidden phase rotation was written:

$$\begin{aligned} v(t) &= R(-\omega t) v(0) \\ R(-\omega t) &= \cos(\omega t) I - \sin(\omega t) J \end{aligned}$$

Differentiating gives the first-order real equation:

$$dv/dt = -\omega Jv$$

Using the complex shorthand $v = (u, w) \leftrightarrow \psi = u + i w$:

$$\begin{aligned} d\psi/dt &= -i \omega \psi \\ i d\psi/dt &= \omega \psi \end{aligned}$$

With $E = \hbar \omega$:

$$i \hbar d\psi/dt = E \psi$$

This is the Schrodinger-type energy-eigenstate form. In this representation, quantum time evolution is the compact notation for hidden transverse temporal rotation.

3B does not derive the physical Hamiltonian H. It shows structural equivalence between unitary phase evolution and real hidden rotation.

6. Toy Model 3C - Spatial waves and kinetic structure

Toy Model 3C introduced spatial dependence. The phase field was:

$$\begin{aligned} \phi(x, t) &= kx - \omega t \\ \psi(x, t) &= \exp(i(kx - \omega t)) \end{aligned}$$

The first spatial derivative gives the momentum structure:

$$\begin{aligned} d\psi/dx &= i k \psi \\ \hat{p} &= -i \hbar d/dx \\ \hat{p} \psi &= \hbar k \psi \\ p &= \hbar k \end{aligned}$$

The second derivative gives the kinetic structure:

$$\begin{aligned} d^2\psi/dx^2 &= -k^2 \psi \\ \hat{T} &= -(\hbar^2 / 2m) d^2/dx^2 \\ \hat{T} \psi &= (\hbar^2 k^2 / 2m) \psi \end{aligned}$$

When the standard nonrelativistic free-particle dispersion relation is imposed:

$$\hbar \omega = \hbar^2 k^2 / 2m$$

the free-particle Schrodinger structure is recovered:

$$i \hbar \frac{d\psi}{dt} = -(\hbar^2 / 2m) \frac{d^2\psi}{dx^2}$$

3C does not derive mass m, $\hbar$, or the dispersion relation. It shows that the kinetic operator structure matches spatial hidden phase twisting.

7. Toy Model 3D - Scalar potentials as local phase-rate shifts

Toy Model 3D added scalar potential energy. From 3B, energy corresponds to phase-rotation rate:

$$E = \hbar \omega$$

Therefore a potential energy $V(x)$ corresponds to a local additional phase-rotation rate:

$$\omega_V(x) = V(x) / \hbar$$

Potential-only evolution becomes:

$$\frac{d\psi}{dt} = -i [V(x)/\hbar] \psi$$

$$i \hbar \frac{d\psi}{dt} = V(x) \psi$$

Combining this with the kinetic structure from 3C gives:

$$i \hbar \frac{d\psi}{dt} = [-(\hbar^2/2m) \frac{d^2}{dx^2} + V(x)] \psi$$

If $V(x)$ is real, the potential operator is Hermitian, the total Hamiltonian is Hermitian, and closed-system norm is preserved.

3D does not derive actual force fields. It shows that scalar potentials are structurally compatible with position-dependent hidden phase-rate shifts.

8. Toy Model 3E - Gauge phase and electromagnetic coupling

Toy Model 3E tested local phase freedom. A local phase transformation is:

$$\psi' = \exp(i q \chi(x,t)/\hbar) \psi$$

The ordinary derivative fails because:

$$\frac{d\psi'}{dx} = \exp(i q \chi/\hbar) \left[\frac{d\psi}{dx} + i(q/\hbar) (d\chi/dx) \psi \right]$$

The fix is to introduce connection fields and covariant derivatives:

$$D_x = d/dx - i(q/\hbar) A$$

$$D_t = d/dt + i(q/\hbar) \Phi$$

$$A' = A + d\chi/dx$$

$$\Phi' = \Phi - d\chi/dt$$

Then the derivatives transform covariantly:

$$D'_x \psi' = \exp(i q \chi/\hbar) D_x \psi$$

$$D'_t \psi' = \exp(i q \chi/\hbar) D_t \psi$$

The covariant Schrodinger structure expands to the minimal-coupling form:

$$i \hbar \frac{d\psi}{dt} = \left[(1/2m) (-i \hbar \frac{d}{dx} - qA)^2 + q \Phi \right] \psi$$

3E does not derive Maxwell equations, electric charge q, QED, or the Standard Model. It shows the structural role of gauge potentials as local phase-connection fields.

9. Toy Model 3F - Relativistic Klein-Gordon / Dirac bridge

Toy Model 3F replaced the nonrelativistic dispersion relation with the relativistic invariant:

$$E^2 = p^2 c^2 + m^2 c^4$$

Using the phase-wave operator substitutions:

$$E \rightarrow i \hbar \frac{d}{dt}$$

$$p \rightarrow -i \hbar \text{grad}$$

gives the Klein-Gordon structural form:

$$[(1/c^2)d^2/dt^2 - \text{grad}^2 + (mc/\hbar)^2] \psi = 0$$

To preserve first-order time evolution, Dirac factorization is required:

$$H = c \alpha \cdot p + \beta m c^2$$

The matrices must satisfy anti-commutation rules:

$$\begin{aligned} \alpha_i^2 &= I \\ \beta^2 &= I \\ \{\alpha_i, \alpha_j\} &= 0 \text{ for } i \neq j \\ \{\alpha_i, \beta\} &= 0 \end{aligned}$$

This produces the Dirac structural equation:

$$i \hbar \frac{d\psi}{dt} = [c \alpha \cdot p + \beta m c^2] \psi$$

A scalar amplitude is no longer enough. The relativistic first-order constraint forces matrix coefficients and therefore a multi-component spinor structure.

3F does not derive particle masses, spin values, charge, QFT, or the Standard Model. It shows that first-order relativistic phase evolution requires spinor-like structure.

10. Consolidated mathematical chain

3T hidden temporal geometry

- > 1T observable projection axis
- > hidden transverse 2T phase plane
- > $J^2 = -I$
- > complex amplitudes as real phase-plane vectors
- > uniform phase rotation
- > Schrodinger-type time evolution
- > spatial phase twisting
- > momentum and kinetic operators
- > local phase-rate shifts
- > scalar potentials
- > local phase-reference variation
- > gauge connection fields
- > electromagnetic minimal-coupling structure
- > relativistic invariant constraint
- > Klein-Gordon structure
- > first-order factorization
- > Dirac spinor structure

This is the full Toy Model 3 structural ladder. It is internally coherent as mathematics. Its physical status remains open.

11. Consolidated numerical verification record

Model	Representative numerical result
3A	$J^2 + I$ error = 0.000e+00; norm preservation and complex-phase equivalence at machine precision.
3B	Schrodinger-type phase-rotation equivalence verified; general Hermitian real-generator equivalence error ~1.776e-15.
3C	Momentum and kinetic operator checks verified; Schrodinger residual ~4.217e-05 from finite-difference numerical approximation.
3D	Hamiltonian Hermiticity error = 0.000e+00; max norm drift under evolution ~4.441e-16.
3E	Covariant derivative and gauge-field invariance verified at ~1e-15 scale; ordinary derivative mismatch nonzero as expected.
3F	Gamma Clifford algebra error = 0.000e+00; H^2 invariant error ~1.137e-13; Dirac eigenvalue error ~2.309e-14.

12. What Toy Model 3 achieved

- It built a real-geometric representation of complex amplitudes from a hidden transverse phase plane.
- It represented Schrodinger-type time evolution as hidden phase rotation.
- It mapped spatial phase gradients to momentum structure.
- It mapped second spatial derivatives to kinetic-energy structure.
- It represented scalar potentials as local phase-rotation-rate shifts.
- It represented electromagnetic minimal coupling as a local phase-connection requirement.
- It showed that relativistic first-order evolution requires matrix factorization and spinor structure.
- It remained consistent with standard quantum mathematics at the structural level.

13. What Toy Model 3 did not achieve

- It did not prove physical 3T existence.
- It did not derive the Born rule from first principles.
- It did not derive the numerical value of $\hbar$.
- It did not derive mass, charge, spin values, or the particle spectrum.
- It did not derive Maxwell equations or field source equations.
- It did not derive quantum field theory.
- It did not derive the Standard Model.
- It did not generate new experimental predictions.
- It did not connect this quantum-dynamics track to gravity or the dark sector yet.

14. Formal guardrail consolidation

The correct classification of Toy Model 3 is: structural compatibility and geometric representation. The incorrect classification would be: derivation of quantum mechanics, derivation of the Standard Model, or proof of the 6D framework.

The central scientific risk is overclaiming. The track successfully shows that the proposed projection architecture can host the mathematics of quantum dynamics. It does not yet show that nature uses this architecture.

Therefore any future publication wording should use restrained terms such as: represented, recovered structurally, modeled as, compatible with, and suggests a geometric interpretation. It should avoid stronger claims such as: proved, derived, explained completely, or solved.

15. Remaining hard gaps

1. Born-rule derivation: Toy Model 3 uses squared norms but does not derive probability weighting from deeper constraints.
2. $\hbar$ origin: $\hbar$ is used as the conversion between phase rate and energy; it is not derived.
3. Mass origin: mass appears as a parameter in kinetic and relativistic relations; it is not derived.
4. Charge origin: q appears in gauge coupling; it is not derived.
5. Field source equations: Maxwell equations and field dynamics are not derived.
6. QFT: particle creation, annihilation, vacuum fields, renormalization, and locality are not addressed.
7. Gravity: the relationship between graph/relational geometry and spacetime curvature remains separate from this quantum track.
8. Dark sector: no dark matter or dark energy predictions arise yet.
9. Experimental predictions: the track currently reproduces known structures rather than predicting new measurable deviations.

16. Recommended next major tracks

Option	Purpose	Risk / difficulty
Track 4A - Gravity / metric bridge	Return to the graph-to-geometry work and attempt a metric-tensor or graph-curvature bridge.	High mathematical difficulty; needed for GR connection.
Track 4B - Dark sector interpretation	Test whether hidden relational degrees could mimic dark matter / dark energy phenomenology.	High risk of speculation without equations.
Track 4C - QFT bridge	Extend the phase-dynamics track to fields, creation/annihilation, and relativistic locality.	Very difficult; likely requires external formal review.
Track 4D - Publication consolidation	Prepare a careful paper combining Toy Models 1-3 with strict claims and roadmap.	Lower risk; useful for communication and review.

17. Close-out statement

Toy Model 3 is closed as a successful structural compatibility chapter. It demonstrates that the reciprocal projection framework can represent the main quantum-dynamics structures from complex amplitudes through relativistic spinor form. It does not establish the framework as physical theory. The next work must either connect this structure to gravity/dark-sector questions or consolidate the results into a publication-grade roadmap.

18. Rebind prompt for continuation

We are resuming after Toy Model 3G.

Toy Model 3G closed the quantum-dynamics track.

Toy Model 3A:

Complex amplitudes can be represented as real hidden transverse phase-plane vectors.

$J = [[0, -1], [1, 0]]$, $J^2 = -I$.

Toy Model 3B:

Uniform hidden phase rotation gives Schrodinger-type time evolution.

$d\psi/dt = -i\omega J\psi$

$i\hbar d\psi/dt = E\psi$

Toy Model 3C:

Spatial phase twisting gives momentum and kinetic structure.

$\hat{p} = -i\hbar d/dx$

$\hat{T} = -(\hbar^2/2m)d^2/dx^2$

Toy Model 3D:

Scalar potential $V(x)$ is a local hidden phase-rotation-rate shift.

$\omega_V(x) = V(x)/\hbar$

Toy Model 3E:

Local phase freedom requires connection fields.

$D_x = d/dx - i(q/\hbar)A$

$D_t = d/dt + i(q/\hbar)\Phi$

This recovers electromagnetic minimal coupling structurally.

Toy Model 3F:

Relativistic constraint $E^2 = p^2 c^2 + m^2 c^4$ gives Klein-Gordon.

First-order factorization gives Dirac structure and forces spinors.

Toy Model 3G conclusion:

The 3T $\rightarrow$ 1T reciprocal projection architecture is structurally compatible with standard quantum-dynamics mathematics, but does not derive the Standard Model, QFT, $\hbar$, mass, charge, the Born rule, or physical 3T existence.

Recommended next major choice:

Either return to gravity/metric/dark-sector bridges, or prepare a publication-grade consolidated paper with strict guardrails.